

Aadarsha Gopala Reddy

+1 (740) 802-1776 | agr@agreddy.com | linkedin.com/in/agopalareddy | github.com/agopalareddy | agreddy.com

EDUCATION

Washington University in St. Louis <i>M.S. Computer Science</i> • Thesis: Gopala Reddy, Aadarsha, “Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data” (2026). <i>McKelvey School of Engineering Graduate Student Theses & Dissertations</i>. 1341. https://doi.org/10.7936/g95n-fv35	St. Louis, Missouri, USA August 2024 - May 2026
Ohio Wesleyan University <i>B.A. Computer Science & Data Analytics</i> • Minor: Economics	Delaware, Ohio, USA August 2020 - May 2023

RESEARCH EXPERIENCE

Washington University in St. Louis <i>Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data</i> DRIVES Project (PI: Dr. Ganesh Babulal; Advisors: Dr. Alvitta Ottley, Dr. Nathan Jacobs) • Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving vehicle-agnostic weekly behavioral features and integrating demographic covariates for CDR status prediction. • Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-participant-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).	St. Louis, Missouri, USA August 2025 - April 2026
Ohio Wesleyan University <i>Opinion Survey on Artificial Intelligence in the Workplace</i> Independent Study (Mentor: Dr. Nicholas Dietrich) • Designed, fielded, and investigated a survey experiment on the impact of AI on 250 employees within each of four industries using survey data. • Analyzed data using Tableau dashboards and R to identify trends and common opinions. • Presented preliminary results on AI’s positive impact on employee work at the OWU Spring Student Symposium. <i>Artificial Intelligence in Modern Board Games — Lost Cities AI</i> Summer Science Research Program (Mentor: Dr. Sean McCulloch) • Developed a digital version of the Lost Cities card game using Java. • Implemented an intelligent agent that played the game against a human, winning 13 of 18 games in testing.	Delaware, Ohio, USA January 2023 - May 2023 May 2022 - July 2022

TEACHING & MENTORSHIP EXPERIENCE

Washington University in St. Louis <i>Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale</i> • Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing strategies, and query optimization. • Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints. • Mentored three capstone teams from raw open datasets to production-grade dashboards, providing weekly feedback that kept all teams on schedule through final presentations. • Held weekly office hours to debug student pipeline implementations across Snowflake schema design, Airflow DAG failures, and Kafka consumer configuration. <i>HackWashU Databases Workshop</i> • Designed and facilitated a hands-on databases workshop for beginner-to-intermediate developers, teaching through two complete projects rather than lectures. • Guided participants through building a normalized SQLite database from the iTunes Search API and a full-stack Supabase Todo app with Row Level Security — covering relational databases, APIs, cloud Postgres, and access control. <i>Engineering Workshop by AGES & Science Coach</i> • Delivered an introductory workshop on artificial intelligence to high school students, covering core AI concepts with hands-on demonstrations.	St. Louis, Missouri, USA January 2026 - May 2026 September 2025 September 2024
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- Tutored approximately ten students for over 6 hours per week, assisting with homework and exam preparation for introductory, intermediate, and advanced computer science coursework.
- Tested and graded assignments of around 35 students in introductory and intermediate computer science classes.

PRESENTATIONS

Ohio Wesleyan University	Delaware, Ohio, USA
<i>Spring Student Symposium</i>	April 2023
Artificial Intelligence Opinion Survey	
<i>Patricia Belt Conrades Summer Science Research Symposium</i>	September 2022
Artificial Intelligence in Modern Board Games	

GRANTS, FELLOWSHIPS, & AWARDS

Bauer Leaders Academy (BLA) Leadership Development Badge , Washington University in St. Louis	April 2026
Affiliated School Scholarship , Washington University in St. Louis	May 2024
Dean's List , Ohio Wesleyan University	May 2023
Mortar Board National College Senior Honor Society Membership , Ohio Wesleyan University	April 2023
Golden Bishop Award - WCSA Best New Member , Ohio Wesleyan University	April 2023
Dean's List , Ohio Wesleyan University	May 2022
Florence Leas Sophomore Prize , Ohio Wesleyan University	April 2022
First Place - 32nd Annual Spring Programming Contest , Denison University	March 2022
International Baccalaureate Scholarship , Ohio Wesleyan University	January 2020
Scholar , Next Genius Scholarship Foundation	January 2020

PROFESSIONAL SERVICE & LEADERSHIP

Washington University in St. Louis	St. Louis, Missouri, USA
<i>Vice President of the Graduate-Professional Council (GPC) Chamber, Graduate and Professional Student Council (GPSC)</i>	May 2025 - May 2026
• Chaired the Constitution Committee (August 2025 - February 2026), coordinating stakeholder input to draft and ratify GPSC governing documents aligned with the Council's mission and university policies. • Built committee membership rosters using ranked-choice preference allocation; represented the graduate student body to university leadership and supported the GPC President in advancing strategic council priorities.	
<i>Member, Graduate Student Affairs Advisory Board (GSAAB)</i>	August 2025 - May 2026
• Provided strategic feedback to the Vice Chancellor for Student Affairs and senior university leaders on critical issues impacting graduate student life and experience. • Contributed to monthly discussions on student affairs policies and initiatives to enhance support systems and resources for the graduate student community.	
<i>Graduate Student Member, Center for Career Engagement (CCE) Student Advisory Board</i>	August 2025 - May 2026
• Guided the strategic direction of the Center for Career Engagement to ensure services align with diverse needs of graduate students from all academic backgrounds and career interests. • Built community around career development initiatives and fostered collaboration between the CCE and graduate student body.	
<i>Treasurer, Umang (Indian Graduate Student Association)</i>	June 2025 - October 2025
• Managed club finances and organized cultural events and festivals, promoting cultural awareness and fostering community among Indian graduate students.	
<i>Treasurer, HackWashU</i>	December 2024 - July 2025
• Managed the club's finances, including budgeting and expense tracking; coordinated fundraising efforts and sponsorship outreach to support hackathons and coding events.	
Ohio Wesleyan University	Delaware, Ohio, USA
<i>Budget Committee Senator, Wesleyan Council on Student Affairs (WCSA)</i>	August 2022 - May 2023
• Proposed and voted on senate bills while overseeing allocation of an annual budget exceeding \$350,000 to campus individuals and organizations; received the Golden Bishop Award for WCSA Best New Member.	

<i>Vice President of Finance, Campus Programming Board</i>	January 2022 - December 2022
Managed annual budgets exceeding \$86,000 for campus-wide programming and secured \$10,000 in additional annual funding through negotiation with WCSA.	

PROJECTS

Washington University in St. Louis	St. Louis, Missouri, USA
<i>Red-Blue Visual Auto Defender: Automated Visual Jailbreak Generation and Explainable Defenses</i>	August 2025 - December 2025
- Built an automated red-blue teaming pipeline for vision-language model (VLM) security, addressing visual prompt injection (VPI) attacks against VLM-based agents. - Generated visual jailbreak attack images by overlaying malicious instructions onto benign images and evaluated them against a target VLM (Gemma-3-4b-it) using semantic response analysis. - Auto-generated deterministic, OCR-based Python defenses for each successful attack, validated with accuracy, precision, and recall metrics in an iterative refinement loop.	
<i>Multimodal Prediction of Alzheimer’s Disease</i>	August 2024 - December 2024
- Developed a multimodal fusion approach for early detection of Alzheimer’s Disease, combining MRI imaging and clinical/demographic data from the OASIS-1 dataset. - Implemented CNNs using TensorFlow/Keras for brain MRI analysis alongside XGBoost and Random Forest models for tabular clinical data. - Created a combined classifier integrating both modalities and documented the project in NeurIPS format with comprehensive performance evaluation metrics and visualizations.	
<i>Datacenter Cooling Optimization using Deep Reinforcement Learning</i>	August 2024 - December 2024
- Implemented multiple deep reinforcement learning algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations to optimize datacenter cooling systems. - Achieved 35.8% energy efficiency improvement over baseline using DDQN, addressing the underserved small to mid-sized datacenter market. - Created a comprehensive framework integrating building energy simulation with reinforcement learning for dynamic thermal management.	
MITxSureStart FutureMakers Create-a-Thon Program	Remote
<i>Parkinson’s Disease AI Diagnosis Software</i>	June 2022 - July 2022
- Built a dual-input pipeline combining audio spectrograms (custom CNN for vocal tremor detection) and spiral-drawing images (Random Forest classifier with augmentation for micrographia detection). - Implemented a weighted ensemble combining both modalities into a single Parkinson’s diagnosis score. - Designed a Flask web interface for user submissions of audio and drawing samples for real-time risk assessment.	

OTHER PROFESSIONAL EXPERIENCE

Washington University in St. Louis	St. Louis, Missouri, USA
<i>Graduate Assistant, Taylor Family Center for Student Success</i>	August 2024 - May 2026
- Planned and ran three recurring engagement programs for first-generation, limited-income (FLI) students — Welcome Dinner, First-Gen Week, and End of Week Unwind — coordinating logistics, promotion, and on-the-day operations. - Authored FERPA-compliant process manuals for engagement data collection and evaluation, giving center staff a standardized, repeatable framework for measuring FLI student participation. - Analyzed semester survey responses and event attendance data, producing reports that directly informed the following cycle’s programming decisions and outreach priorities. - Managed the center’s weekly newsletter; represented the department at off-campus excursions and FLI student trips.	

Crittero, Inc.

Remote

AI Engineer Intern

June 2025 - August 2025

- Generated 100,000 labeled social media posts across seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model achieving 80% accuracy (30-percentage-point gain over random baseline) with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.
- Built a configurable Python CLI test harness (700+ lines) with deterministic seeding and CSV export for reproducible model evaluation, using it to validate accuracy across manually tuned embedding-dimension and cold-start-threshold configurations.

Lab714

Boca Raton, Florida, USA

Data Analyst Intern

June 2023 - May 2024

- Analyzed IoT environmental sensor data streams on AWS, building dashboards that surfaced chemical imbalances and equipment anomalies through automated threshold detection and trend analysis.
- Built AWS data ingestion pipelines to normalize and store continuous sensor telemetry from deployed devices, supporting real-time operational monitoring of deployed IoT fleets.
- Wrote SQL queries to aggregate historical sensor data by pool, date range, and chemical parameter, powering the trend analytics and anomaly detection shown in client dashboards.
- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++, R, C#, Rust.

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, React, Vue.js, Node.js, Express.js, FastAPI, Socket.IO, Streamlit.

Tools & Technologies: L^AT_EX, Git, Tableau, Power BI, AWS, Docker, PostgreSQL, MySQL, MongoDB, Snowflake, Apache Airflow, Spark, Kafka, Flink, Gemini/OpenAI APIs, Microsoft 365, Google Workspace.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.